LEVELHULL · PROBLEM STATEMENT

LevelHull problem statement

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DOCUMENT CONTROL

Document ID	LVH-PRB-001	Version	0.4
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/01-problem.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-24	Amish Chadha	Users and needs, and stakeholders tables added (kit 1.8 pipeline, STANDARDS section 22); they are the sources of LVH-URS-001
0.3	2026-10-03	Amish Chadha	Settled answers restated for the 220 mm modules (LVH-DDR-003)
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 update; constraints restated, open questions settled (LVH-DDR-001), first co-design candidates, safety section
0.1	2026-09-30	Amish Chadha	Initial scaffold

LevelHull problem statement

When a wooden fishing canoe swamps, the crew loses the boat as a refuge just when they need it most. A small amount of well-placed buoyancy could change that.

The problem

Deaths cluster in bad weather and at night, when waves swamp boats and rescue is least likely (The Conversation). Life jackets are the first line of defence, but many fishers do not wear them. A boat with enough built-in buoyancy to float level when swamped gives a second chance, which is what the US level flotation test checks (USCG).

Existing answers do not fit these boats. The US test is a benchmark for boats, not a method for retrofitting wooden fishing canoes. Inflatable emergency floats exist but depend on an inflation mechanism, and at least one such design is under patent until about 2032 (US9139267B2). What is missing is an open, passive retrofit kit, with a sizing method, that a local builder can fit to the boats fishers already own.

Users and context

Table 1. Users.

User	Need	Context
Canoe fishers and crews	A boat that stays afloat and level when swamped, so they can hold on and bail	Lakes, rivers and coasts; storms and night fishing; often no life jacket
Canoe owners	A retrofit that is affordable, lasts, and takes little space from the catch	Wooden canoes of many sizes, often with outboard motors
Local boatbuilders	A sizing sheet and fitting guide for different hulls	Beach yards with hand tools
Fisheries departments, beach management units and NGOs	A measure to promote alongside life jackets and weather warnings	Safety at sea programmes

Table 1a. Users and needs (kit 1.8; the source of the user requirements in LVH-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Canoe fishers and crews	A canoe that stays afloat, upright and level when it fills, something to hold, and a way to bail it out and get home	Four crew setting or hauling nets at night or in a squall; a wave fills the canoe; they hold on, then bail	Many cannot swim and many wear no life jacket; cold, darkness and exhaustion; crowding on one side
Canoe owners	A retrofit that is affordable, lasts years in sun, fuel and water, keeps room for catch and nets, and does not hinder beaching or landing	One owner, several crews; the canoe is dragged up the beach daily and rafted alongside others at the landing	Cash flow; space is income; a kit that rots, tears or gets in the way will be thrown out
Local boatbuilders	A sizing method and a fitting guide that work on the hulls they build, with hand tools and local materials	Fitting the kit to a canoe on the beach in a day, between repairs	Every hull differs; no drawings; no welding or fusion equipment; holes below the waterline are a risk to their reputation
Fisheries departments, beach management units and NGOs	A measure to promote with life jackets and weather warnings that saves lives and does not make crews careless	Safety campaigns at landing sites; licensing and inspection of canoes	Fear that flotation is used as an excuse not to wear life jackets; approval of changes to boats

Table 1b. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Beach management units (BMUs)	Run the landing sites; trusted by fishers; enforce safety rules	Co-design partner for the first fitting and swamp test
National fisheries research institutes	Hull, load and catch surveys; data on accidents	Hull measurements and loads for the sizing method
Maritime authorities (for example the Kenya Maritime Authority)	Small vessel safety rules, approval of modifications	Ask whether fitting needs approval (LVH-REG-001)
Sea-safety and life jacket programmes, NGOs	Training fishers; promotion	Promote the kit alongside life jackets, never instead
Foam, tarpaulin and webbing suppliers	Materials	Data sheets for foam density, water uptake and UV life
Open hardware community	Builders elsewhere	Feedback on the sizing method for other hulls

Operating environment

- Fresh and salt water; lakes, estuaries and near-shore coastal waters.
- Strong sun and heat; boats beached and left in the open.
- Wooden hulls that take on water and are often wet inside.
- Fuel, fish slime, nets and abrasion from gear.
- Loads of crew, catch, gear and sometimes an outboard motor.

Constraints

- Value-engineering target: USD 1,000 for the kits for two prototype canoes (a hypothetical control target, not a spending limit; STANDARDS section 18).
- Passive only: closed-cell foam or sealed rigid chambers; no inflation.
- Must not take up so much volume that it blocks work or catch storage.
- Fitted with hand tools by a local boatbuilder, without changing the hull shape and without new holes below the waterline.
- Materials must resist UV, fuel and abrasion and must not soak up water.
- Open design: hardware under CERN-OHL-S-2.0, sizing calculator under MIT.

Out of scope

- Life jackets and personal flotation.
- New hull designs.
- Engines, propulsion and navigation equipment.
- Certification as lifesaving equipment.

Prior work

Table 2. Prior work.

Prior work	What it does	Gap for these users	Source
US Coast Guard level flotation test	Test method checking that a swamped small boat floats approximately level	A test benchmark, not a retrofit method or kit for wooden fishing canoes	link
US9139267B2, inflatable float (active to about 2032)	Emergency float that inflates to keep a craft afloat	Active inflation mechanism and patent restrictions; LevelHull stays passive	link
RNLI sustainable lifesaving equipment	Programme supporting locally made lifesaving equipment in lower-income countries	Not a boat flotation retrofit	link

Co-design

The first candidates to approach, in order (none approached yet): a beach management unit on the Kenyan shore of Lake Victoria, working with the canoe builders at its landing site; a national fisheries research institute in the region, for hull and load surveys; and a sea-safety programme that already trains fishers in the use of life jackets, so the kit is promoted alongside them. With the partner, measure real hulls (frame spacing and depth, flare, timber, motor) and loads, and swamp-test fitted canoes in sheltered, shallow water.

Questions settled at TRL 2

These were the open questions of v0.1. Each is decided in LVH-DDR-001 under Amish's pre-approval of 2026-10-03; the numbers are in LVH-CAL-001.

Table 3. Questions settled.

Question	Answer
Low along the sides, or higher under the gunwales?	High, under the gunwales, so the modules cut the waterline when the canoe is swamped. The same foam fitted low on the floor only just reaches the water surface: with four crew on one side it heels past 10 deg, and a few kilograms more load leave it with no roll stiffness at all, against 761 kg m per radian with the modules high.
Foam or sealed chambers?	Closed-cell polyethylene foam in tarpaulin covers. A puncture loses no lift and nothing needs sealing; chambers are kept for the sibling LaneSkiff concept.
How much lift does the wood give, and what does the motor change?	The design takes no credit for the wood: wet hull timber is counted as exactly as dense as water. A typical 750 kg/m3 hardwood would add about 54 mm of freeboard. The motor is counted at its full 36 kg at the transom; other motors are sized with the calculator.
Re-entry step?	Still parked; scope unchanged. Grab lines along both sides give the crew something to hold while they are in the water.
Promotion without implying life jackets are optional	Every document, the fitting guide and the kit itself carry "never a substitute for life jackets"; the swamp test is run with every person in a life jacket.

Safety

> **Safety:** LevelHull is flotation for a boat, not lifesaving equipment for people. It is never a substitute for life jackets, and it is not certified. A swamped canoe that floats can still roll in breaking waves and cannot protect the crew from cold water, night or exhaustion; the kit buys time to bail and to be seen. Fitted canoes are swamp-tested in sheltered, shallow water with a safety boat and every person in a life jacket before they go fishing.